

Total No. of Questions : 10]

SEAT No. :

P3051

[Total No. of Pages : 2

[5354]-539

B.E. (Mechanical) (Semester - I)

PRODUCT DESIGN AND DEVELOPMENT(Theory)
(2012 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:-

- 1) *All questions are compulsory. i.e. Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.*
- 2) *Figure to the right indicate full marks.*

Q1) Discuss in detail various factors affecting the product design? Enlist various basic planning and scheduling Tools? **[10]**

OR

Q2) a) Discuss various criterion to define Market Segments? **[4]**

b) What is Customer? Explain concept of Average Customer & 3-Sigma Customer. **[6]**

Q3) Explain Product Function analysis with FAST method. Elaborate with suitable product example. **[10]**

OR

Q4) What is Technology Forecasting? Explain the concept with any one tool? State Importance. **[10]**

Q5) a) Explain following tools used in Benchmarking Process: **[8]**

i) Intended Assembly Cost Analysis

ii) Trend Analysis

b) Discuss Step wise procedure of Product Teardown Process in Detail? **[8]**

P.T.O.

OR

- Q6)** a) Apply Subtract and Operate procedure for paper punching machine. Analyse & Explain Force flow in machine with suitable diagram. [8]
- b) Explain in detail the process of Setting Product Specifications with suitable Example? [8]

- Q7)** a) Explain Manufacturing Cost Analysis. Explain stepwise procedure for estimation of cost. [8]
- b) Discuss Guidelines of Design for Casting & Forging. [8]

OR

- Q8)** a) Discuss Guidelines for Design for Environment and Materials in detail. [8]
- b) Explain Guide lines of Design for Robustness and Safety. [8]

- Q9)** a) Explain Product Workflow and Product Data Management System in detail. [9]
- b) Explain the role of customers and vendors in implementation of Product Life Cycle Management. [9]

OR

- Q10)** a) Discuss Product, Processes, People and methods as components of Product Life Cycle Management. [9]
- b) Draw typical Product Life Cycle and Explain Phases? [9]

